

Josiah Holder

Mechanical Engineer

Fargo, ND | josiahholder523@gmail.com | linkedin.com/in/josiahholder523 | josiah05230.github.io

SUMMARY

Mechanical engineer with three and a half years of product development experience in agricultural equipment. Has taken sensor designs from prototype to production, including CAD, drawing and tolerance control, supplier sourcing, and design validation. Works regularly with farmers, dealers, and manufacturing partners to turn field feedback into design improvements.

SKILLS

CAD and drawings: Creo, Fusion 360, GD&T, tolerance stack-up analysis, design for manufacturability, revision control

Manufacturing: Injection molding, SLS, machining, sheet cutting, overmolding, supplier quoting, end-of-line testing

Prototyping and testing: FDM and SLS 3D printing, test stand design, fluid and pneumatic test equipment, thermal chamber testing, data acquisition, CPK and PPK studies

Software and analysis: Python, Julia, C#, Git, statistical analysis, introductory CFD

EXPERIENCE

Precision Planting: R&D Engineer

January 2025 - Present, Fargo, ND

- Designed three mechanical concepts for an optical seed-flow sensor using injection molding, SLS, machining, and sheet cutting; obtained quotes from multiple suppliers and selected the lowest-cost design without sacrificing usability
- Maintain CAD models and production drawings from prototype through release, applying GD&T (flatness, surface finish, ordinate dimensioning) and tolerance stack-up analysis
- Reduced sensor-to-sensor variance by roughly 3x through tighter tolerances and assembly design changes
- Developed the test stand, control software, and data analysis used to show that the sensor met its $\pm 2\%$ accuracy requirement, with 95% of units within $\pm 2\%$ of the mean, and increased measurement range by about 80%
- Tested 50+ sensors on the bench and analyzed field data from 250+ units over two growing seasons
- Built fluid and pneumatic test equipment using VFDs, blowers, pumps, flow meters, pressure sensors, and venturi conveying; performed thermal chamber environmental testing
- Designed the mechanical components of an overmolded connector assembly entering production this winter, along with adapter, lab, and field harnesses
- Developed a Python application that creates part views from STEP assemblies for product support, saving an estimated 4+ engineering hours per product
- Meet with farmers and dealers to gather feedback, and work with suppliers and contract manufacturers on cost and end-of-line testing

Appareo Systems (Intelligent Ag): R&D Engineering Intern

March 2023 - January 2025, Fargo, ND

- Recovered a sensor program at risk of cancellation by reworking the mechanical design to perform across a demanding operating range and developing a heat map algorithm that improved accuracy from roughly $\pm 60\text{-}80\%$ to $\pm 5\%$
- Eliminated a calibration step that skewed the reference measurement, and accounted for variation between product types
- Planned and ran lab and field testing, built test stands, and analyzed data, carrying the responsibilities of a full-time engineer

Faul Building and Repair: Foreman

Summer 2021, Fargo, ND

- Supervised a crew of up to 8 on demolition, framing, and repair projects; managed scheduling and served as the main point of contact for the owner and customers

Co-founder, small business

2024 - Present, Fargo, ND

- Building low-cost, easy-to-use technical solutions for farmers

EDUCATION

North Dakota State University: B.S. Mechanical Engineering

December 2024